

NAYEEM HASAN

SOFTWARE ENGINEER

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SUMMARY

Senior Software Engineer with 8+ years of experience building secure, high-performance distributed systems across EU healthcare research, fintech, and e-commerce — including platforms achieving 2X traffic capacity with architecture built for 4X peak load. Currently leading backend platform development for an active EU federated learning research project (FLUTE), with successfully delivered infrastructure for a completed EU project (TRUMPET). Strong expertise in Python (FastAPI, Django), PHP (Laravel), Node.js (Express.js), and cloud-native deployments on Azure and AWS.

PROFESSIONAL EXPERIENCE

Technovative Solutions LTD, Senior Software Engineer

May 2024 - Present

Tech stack: Python, FastAPI, Django, Celery, PostgreSQL, Docker, SPIRE, Envoy, Azure, GitHub, Pytest

- Led backend platform development for two EU federated learning research projects (FLUTE & TRUMPET), building cloud platforms and on-premises Data Hub/Data Owner Nodes that enable privacy-preserving AI workflows across distributed hospital partners without raw clinical data leaving local premises.
- Built role-based dashboards and RESTful APIs serving Researchers, Data Hub operators, and Governance/Admin roles across both platforms, with audit-ready access control, study approval workflows, and asynchronous task processing using Celery.
- Integrated HL7 FHIR and DICOM/Orthanc multimodal data pipelines standardising clinical datasets across hospital partners, supporting federated oncology workflows for NSCLC, HNC, and SBRT use cases.
- Managed decentralised infrastructure security using mTLS with SPIRE/Envoy zero-trust architecture between cloud and on-premises hospital nodes, with Federated Learning Core integration enabling privacy-preserving model training across distributed sites on Azure.

Shajgoj Limited, Software Engineer

Mar 2022 - April 2024

Tech stack: PHP, Laravel, Redis, MySQL, Node.js, MongoDB, Apache Solr, Neo4j, Digital Ocean

- Redeveloped the core e-commerce backend from scratch, achieving 2X current traffic capacity with architecture capable of handling 4X peak load, and reducing average API response time by ~65%.
- Orchestrated migration of 2M+ orders, user records, and transactional data with zero downtime and seamless user journey continuity.
- Implemented a graph-based recommendation engine (Neo4j) and Apache Solr-powered search, improving product discovery and conversion.
- Delivered business-critical discount, cart, and order logic across desktop, mobile, and CRM platforms, while eliminating key system bottlenecks.
- Deployed and optimised cloud infrastructure on Digital Ocean, and implemented event-driven background job processing using Laravel Queues with Redis, reducing system load during peak traffic.

Walletmix Limited, Full Stack Developer

Feb 2021 - Feb 2022

Tech stack: PHP, Laravel, PostgreSQL, Redis, Vue.js, Docker, MySQL, GitHub

- Built a one-stop citizen engagement and service management platform for Dhaka North City Corporation (DNCC), serving 241K+ citizen records across active and in-process service workflows.
- Developed mobile APIs for Robi Alpha subscriber platform, ensuring full backward compatibility across existing client integrations.
- Maintained and extended a homegrown payment gateway platform providing payment infrastructure services to Bangladeshi businesses, handling bug fixes, feature rollouts, and system stability.

Japan Marketing and Consultancy Limited, Full Stack Developer

Jul 2019 - Jan 2021

Tech stack: PHP, Laravel, Vue.js, MySQL, Docker, AWS, Google Cloud, GitHub

- Collaborated across a 6-person cross-cultural development team (Bangladesh and Japan) to deliver a hotel booking platform targeting small-to-medium hotels in the Japanese market unable to access larger platforms.
- Implemented server-side business logic and RESTful APIs supporting hotel registration, search, and booking workflows across desktop and mobile clients.
- Deployed and maintained the system on AWS at launch, optimizing for reliability and operational efficiency.

Orange Business Development Limited, Junior Software Engineer

Oct 2018 - Jun 2019

Tech stack: PHP, Laravel, Vue.js, MySQL, GitHub

- Contributed to an a2i initiative platform streamlining website launches with essential onboarding information.
- Delivered features per project requirements, actively participating in cross-functional project meetings.

Dream 71 Bangladesh Limited, Junior Software Engineer

Jan 2018 - Sep 2018

Tech stack: PHP, Laravel, Vue.js, MySQL

- Developed client-facing software as part of a collaborative backend team, following agile/scrum practices.

KEY PROJECTS

FLUTE — Federated Learning & Multi-Party Computation Platform (EU Research)

- Built the backend platform and API layer enabling privacy-preserving ML workflows across European hospitals.
- Integrated the Federated Learning Core into the platform infrastructure, enabling privacy-preserving model training across distributed hospital Data Hub nodes using PETs developed by research partners.
- Integrated HL7 FHIR pipelines to standardize multimodal clinical datasets (structured records and medical imaging via Orthanc/DICOM) across hospital partners, supporting federated oncology workflows for NSCLC, HNC, and SBRT use cases.
- Managed decentralized infrastructure including mTLS security using SPIRE/Envoy architecture for zero-trust workload communication.

TRUMPET — TRUstworthy Multi-site Privacy Enhancing Technologies (EU Research)

- Built the Cloud platform powering the Researcher, Data Owner, and Governance dashboards, with RESTful APIs, PostgreSQL backend, and role-based access control for multi-institutional federated workflows.
- Integrated the Federated Learning core aggregator into the cloud platform, enabling privacy-preserving model training across hospital Data Owner nodes without raw clinical data ever leaving local premises.
- Designed and implemented the Data Owner Node — an on-prem, privacy-by-design component — covering dataset ingestion, FHIR-based data publishing, study approvals, and federated training execution against locally held clinical data.
- Platform was validated across three real-world clinical use cases (NSCLC, HNC, SBRT) involving hospital partners across Europe.

Shajgoj — E-Commerce Platform

- Led full backend and API development for mobile and CRM applications serving Bangladesh's leading beauty e-commerce brand.
- Built an extensive admin dashboard, recommendation engine, and multi-gateway payment integration.

TECHNICAL SKILLS

Languages: Python, PHP, JavaScript

Frameworks & Libraries: FastAPI, Django, Laravel, Node.js (Express.js), Vue.js

Databases: MySQL, PostgreSQL, MongoDB, Neo4j, Redis

Infrastructure & DevOps: Docker

Architecture: Microservices, REST API, Event-Driven

Cloud Services: AWS, Google Cloud, Digital Ocean, Azure

Security: Zero-Trust Architecture (mTLS, SPIRE, Envoy), OAuth2

Healthcare & Interoperability: HL7 FHIR, DICOM

Testing Tools: Apache JMeter, Pytest

Version Control: Git

EDUCATION

B.S.C in Computer Science and Engineering (CSE)

Jan 2013 - Dec 2017

East West University

LANGUAGES

- **English:** Full Professional Proficiency